Analytic Gaussian Beam Verification of Cavity Propagation

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1 Approach

To verify that the forward and backward propagation steps in the numerical cavity model are implemented correctly, we compare the simulation against the analytic propagation of a gaussian beam. A gaussian beam remains gaussian under both free–space propagation and quadratic phase transformations such as those produced by spherical mirrors.

1.1 The q-parameter

According to Siegman Chapter 20, the gaussian beam parameter $\hat{q}$ transforms as

$$\hat{q}_2 = \frac{A\hat{q}_1 + B}{C\hat{q}_1 + D}, \tag{1}$$

where A, B, C, and D represent the elements of the ray transfer ($ABCD$) matrix associated with the corresponding optical element.

In general,

$$\begin{pmatrix} x_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x_1 \\ \theta_1 \end{pmatrix}. \tag{2}$$

The same matrices that describe ray optics also describe gaussian beam propagation. So once we know the ray matrices for each element, we immediately know how the Gaussian beam evolves and $\hat{q}$ can be used to evolve the beam.

1.2 Gaussian Beam Representation

A Gaussian beam may be written in terms of the beam parameter $\hat{q}(z)$ as

$$E(r, z) = E_0 \frac{q_0}{q(z)} \exp\left(-i\frac{kr^2}{2q(z)}\right) \tag{3}$$

where

$$\frac{1}{q(z)} = \frac{1}{R(z)} - i\frac{\lambda}{\pi w(z)^2} \tag{4}$$

Here

- $w(z)$ is the beam radius
- $R(z)$ is the radius of curvature
- $k = 2\pi/\lambda$

The beam remains Gaussian throughout propagation, and all evolution of the beam is contained in the parameter $\hat{q}$.

1.3 Free Space Propagation

The $ABCD$ matrix for free space propagation is

$$\begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}. \tag{5}$$

In the $\hat{q}$ parameter formulism, propagation through a distance L transforms the beam parameter as

$$\hat{q}_2 = \hat{q}_1 + L \tag{6}$$

Thus if the beam parameter at the left mirror plane (LHS) is

$$q_0 = q_{\text{LHS}} \tag{7}$$

then after propagating to the right mirror plane (RHS)

$$q_{\text{RHS}} = q_0 + L \tag{8}$$

The analytic electric field profile at the RHS plane is therefore

$$E_{\text{RHS}}(r) = E_0 \frac{q_0}{q_{\text{RHS}}} \exp\left(-i\frac{kr^2}{2q_{\text{RHS}}}\right) \tag{9}$$

1.4 Mirror Reflection

The $ABCD$ matrix for mirror reflection is

$$\begin{pmatrix} 1 & 0 \\ -\frac{2}{R_c} & 1 \end{pmatrix}, \tag{10}$$

An infinitely wide spherical mirror of radius of curvature R_c applies a quadratic phase shift

$$E_{after} = E_{before} \exp\left(-i\frac{kr^2}{R_c}\right) \tag{11}$$

which corresponds to the transformation

$$\frac{1}{\hat{q}_2} = \frac{1}{\hat{q}_1} - \frac{2}{R_c} \tag{12}$$

or equivalently

$$\hat{q}_2 = \frac{\hat{q}_1}{1 - \frac{2\hat{q}_1}{R_c}} \tag{13}$$

Thus after reflecting from the RHS mirror of radius R_2

$$q^+_{\mathrm{RHS}} = \frac{q_{\mathrm{RHS}}}{1 - \frac{2q_{\mathrm{RHS}}}{R_2}} \tag{14}$$

The field immediately after reflection is

$$E^+_{\mathrm{RHS}}(r) = E_0 \frac{q_0}{q^+_{\mathrm{RHS}}} \exp\left(-i\frac{kr^2}{2q^+_{\mathrm{RHS}}}\right) \tag{15}$$

1.5 Verification Strategy

The numerical propagation algorithm can be verified by comparing the simulated field profiles with the analytic Gaussian profiles at each stage of the sequence of one round-trip:

1. Propagate from LHS $\rightarrow$ RHS
2. Apply RHS mirror (mirror 1)
3. Propagate from LHS $\rightarrow$ LHS
4. Apply LHS mirror (mirror 2)

2 Results

(a) Numeric solution (b) Analytic solution

Figure 1: Numeric vs. analytic solution for the propagation described above.

2.1 Field Comparison Error Metrics

To quantify the agreement between two fields, several error metrics were computed after first removing the best-fit global complex scaling between the two fields. The best-fit complex scalar α was determined such that

$$E_{\text{num}} \approx \alpha E_{\text{an}}, \tag{16}$$

where

$$\alpha = \frac{\sum E_{\text{an}}^* E_{\text{num}}}{\sum |E_{\text{an}}|^2}. \tag{17}$$

This removes any irrelevant global phase offset and overall amplitude scaling so that the comparison focuses on differences in the spatial phase profile rather than arbitrary normalization.

Complex Field L_2 Error The normalized L_2 error measures the root-mean-square difference between two complex fields:

$$\epsilon_{E,L2} = \frac{\|E_1 - E_2\|_2}{\|E_1\|_2},$$

where E_1 is the reference field (analytic) and E_2 is the field being evaluated (numeric). This metric captures discrepancies in both the amplitude and phase of the optical field and provides a global measure of the error across the computational domain.

Complex Field Maximum Error The maximum pointwise difference between the complex fields is given by

$$\epsilon_{E,\max} = \frac{\max |E_1 - E_2|}{\max |E_1|}$$

This metric highlights the largest local deviation between the two fields.

Intensity L_2 Error Since many physical observables depend on optical intensity rather than the complex field itself, an additional comparison is performed using the intensity distributions

$$I_1 = |E_1|^2, \qquad I_2 = |E_2|^2.$$

The normalized L_2 intensity error is

$$\epsilon_{I,L2} = \frac{\|I_1 - I_2\|_2}{\|I_1\|_2}.$$

This metric evaluates how closely the power distribution of the two fields matches.

Intensity Maximum Error The maximum pointwise difference in intensity is also computed:

$$\epsilon_{I,\max} = \frac{\max |I_1 - I_2|}{\max |I_1|}.$$

Phase Difference Error Metric To quantify agreement between the analytic and numerical complex fields, a phase-difference error metric was computed.

The phase difference at each grid point is then defined as

$$\Delta\phi(x,y) = \arg[\exp(i\,(\phi_{\text{num}}(x,y) - \phi_{\text{ana}}(x,y)))]\,, \tag{18}$$

where

$$\phi_{\text{num}}(x,y) = \arg\big(E_{\text{num}}(x,y)\big), \qquad \phi_{\text{fit}}(x,y) = \arg\big(\alpha E_{\text{an}}(x,y)\big). \tag{19}$$

Writing the phase difference in this wrapped form ensures that the error is confined to the interval $(-\pi, \pi]$, thereby avoiding artificial 2π discontinuities.

Because phase becomes numerically meaningless in regions where the field amplitude is vanishingly small, the phase error is evaluated only over points whose normalized intensity exceeds a prescribed threshold. Defining the numerical intensity as

$$I_{\text{num}}(x,y) = |E_{\text{num}}(x,y)|^2, \tag{20}$$

the mask used for phase evaluation is

$$\frac{I_{\text{num}}(x,y)}{\max\limits_{x,y} I_{\text{num}}(x,y)} > I_{\text{thresh}}, \tag{21}$$

where I_{thresh} is the chosen relative intensity threshold.

The reported phase error is the root-mean-square (RMS) phase difference over all grid points satisfying this mask:

$$\varepsilon_{\phi,\text{RMS}} = \sqrt{\frac{1}{N_m} \sum_{\text{mask}} \Delta\phi(x,y)^2}, \tag{22}$$

where N_m is the number of grid points included in the masked region. A maximum phase-difference metric may also be reported as

$$\varepsilon_{\phi,\max} = \max_{\text{mask}} |\Delta\phi(x,y)|. \tag{23}$$

This metric provides a direct measure of how well the numerical propagation reproduces the analytic quadratic phase curvature of the Gaussian beam while excluding low-intensity regions where the phase is dominated by numerical noise.

All together, these metrics provide a comprehensive assessment of the agreement between two fields by capturing both global and local differences in the complex field and the resulting intensity distribution.

2.2 Field Comparison Error Metrics: Results

- Best-fit $\alpha = 1.27$
- Field L_2 error $= 7.87 \times 10^{-6}$
- Field max error $= 4.21 \times 10^{-6}$
- Intensity L_2 error $= 2.28 \times 10^{-7}$
- Intensity max error $= 1.05 \times 10^{-6}$
- Phase RMS error [rad] $= 2.15 \times 10^{-6}$
- Phase max error [rad] $= 6.75 \times 10^{-6}$
- Normalized field overlap $= 1$